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Chien et al.

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(54) **LIGHT SOURCE SYSTEM**

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F21S 41/20 (2018.01)
F21S 41/25 (2018.01)

(52) **U.S. Cl.**
CPC **F21S 41/176** (2018.01); **F21S 41/25** (2018.01); **F21S 41/285** (2018.01)

(58) **Field of Classification Search**
CPC F21S 41/176; F21S 41/25; F21S 41/285
See application file for complete search history.

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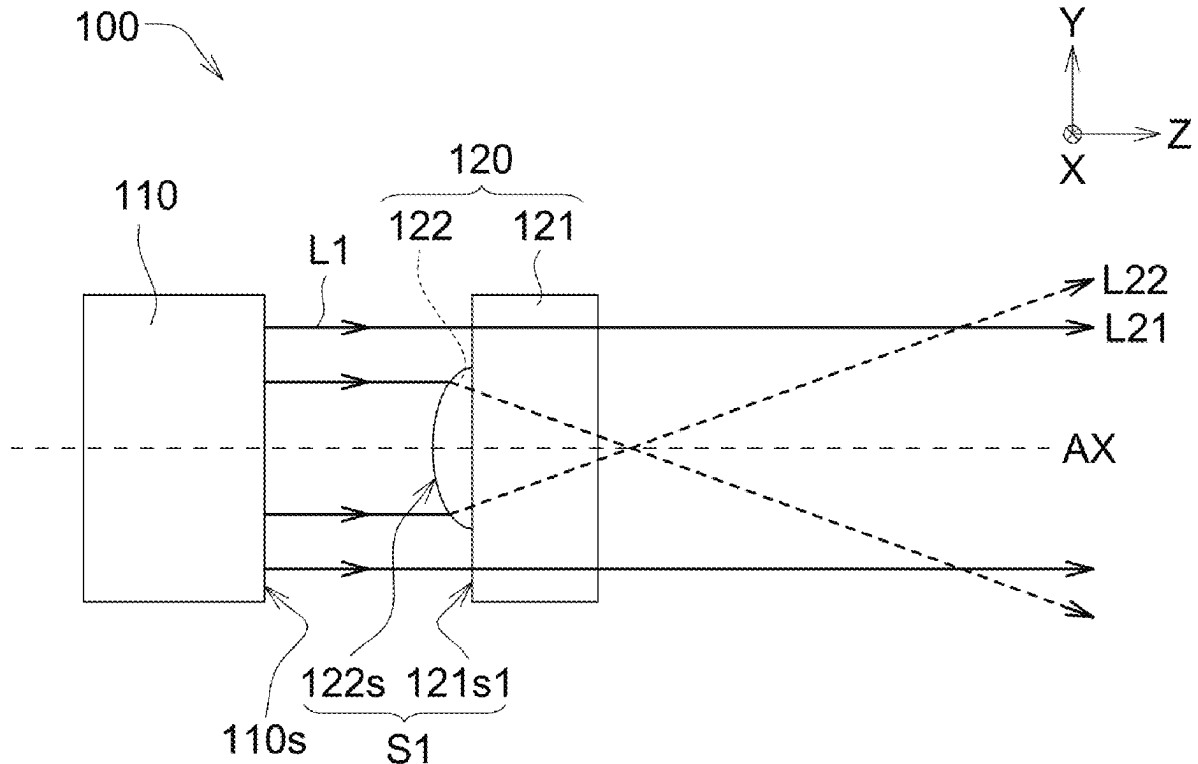
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Primary Examiner — Christopher E Dunay

(57) **ABSTRACT**

A light source system includes a first light source and a beam conversion element. The first light source is configured to emit a first light. The beam conversion element is configured to convert the first light into a condensed light and a diffused light. The beam conversion element has a light transmission surface, the light transmission surface includes a flat region and a curved-surface region, the flat region completely surrounds the curved-surface region, the flat region is configured to convert the first light into the condensed light, and the curved-surface region is configured to convert the first light into the diffused light.

20 Claims, 9 Drawing Sheets



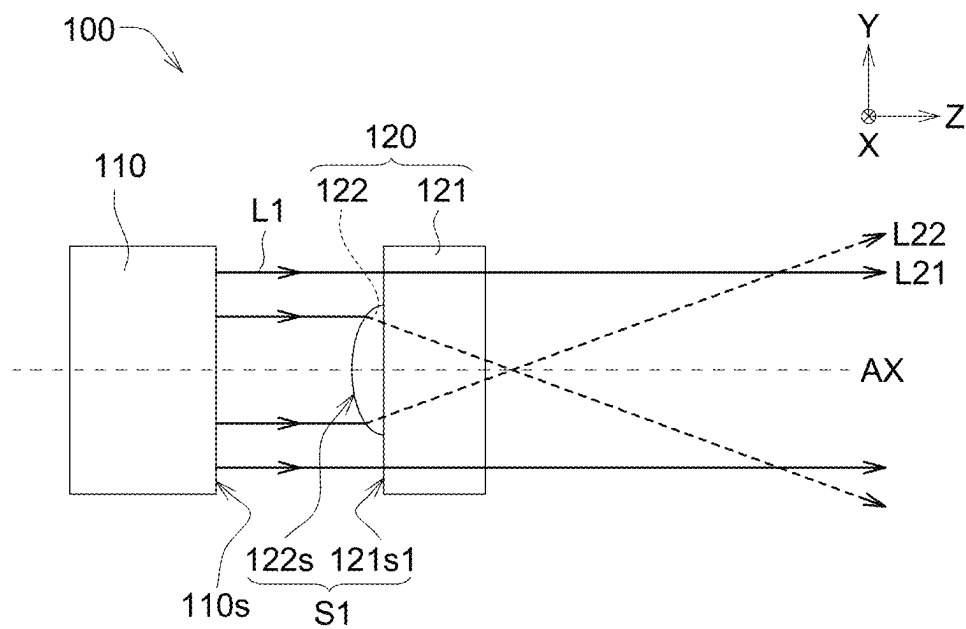


FIG. 1

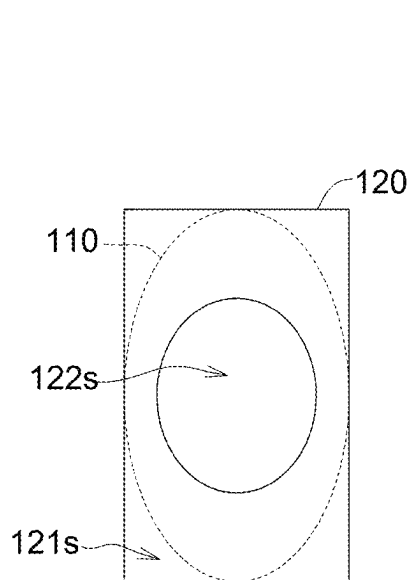


FIG. 2

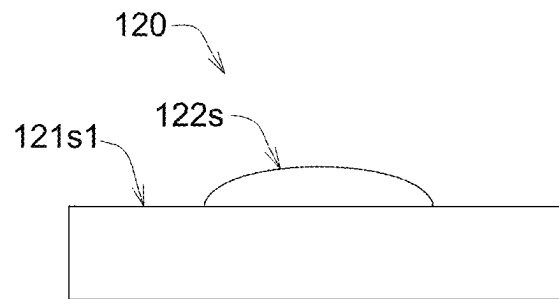


FIG. 3A

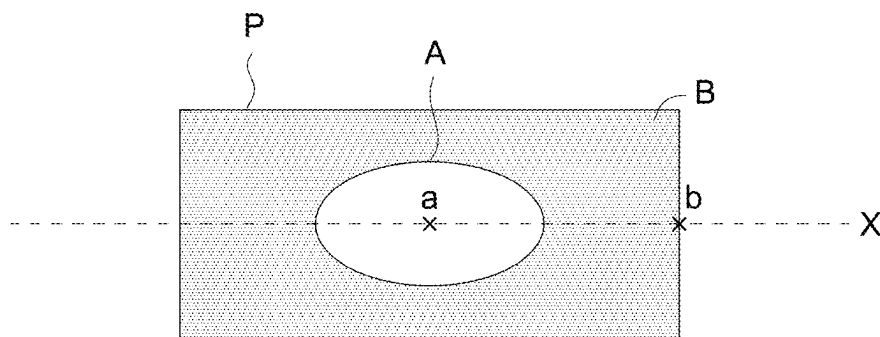


FIG. 3B

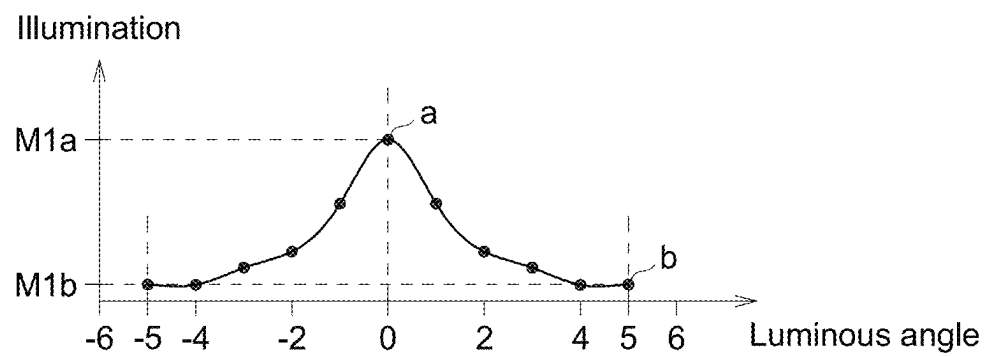


FIG. 3C

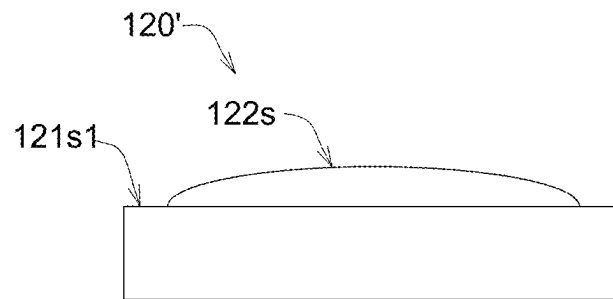


FIG. 4A

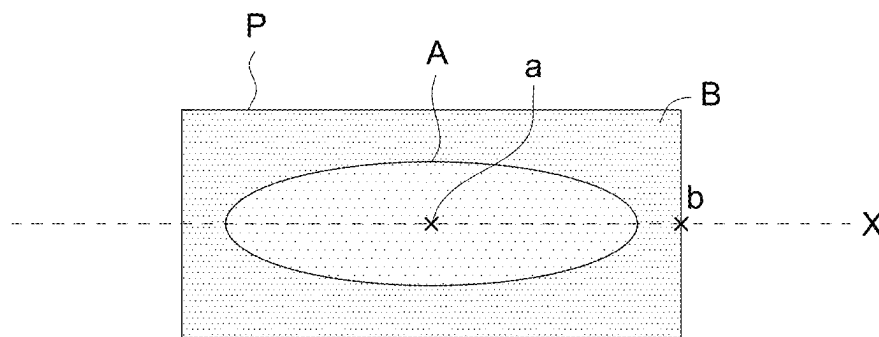


FIG. 4B

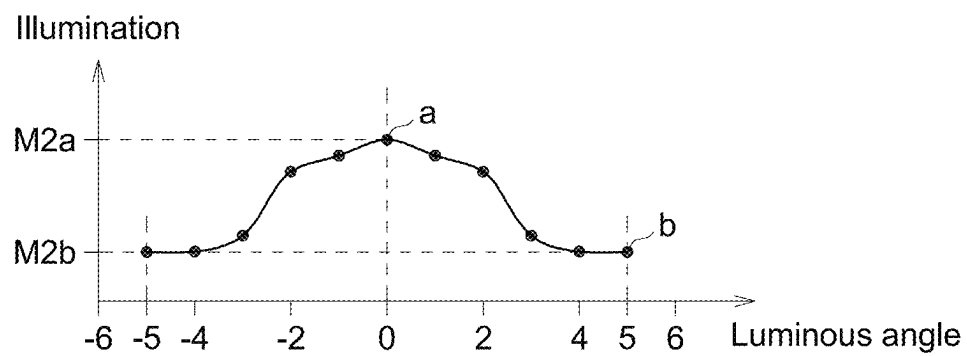


FIG. 4C

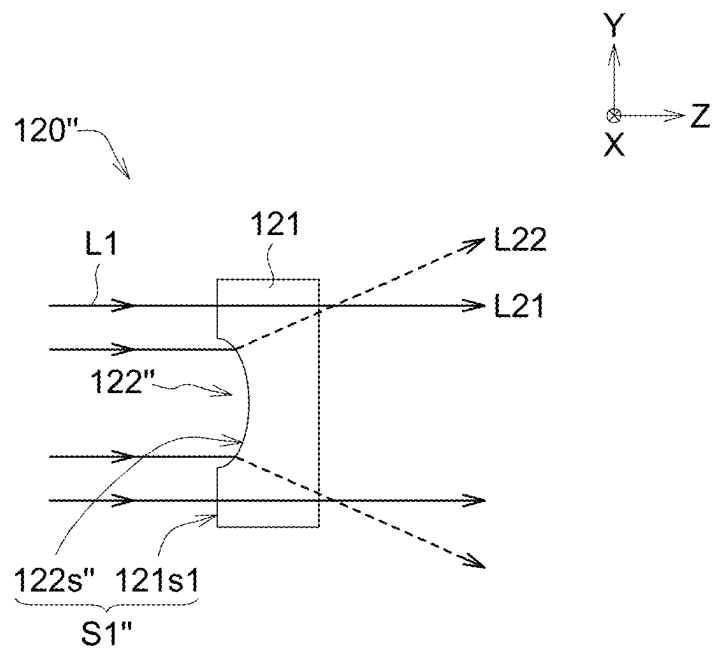


FIG. 5

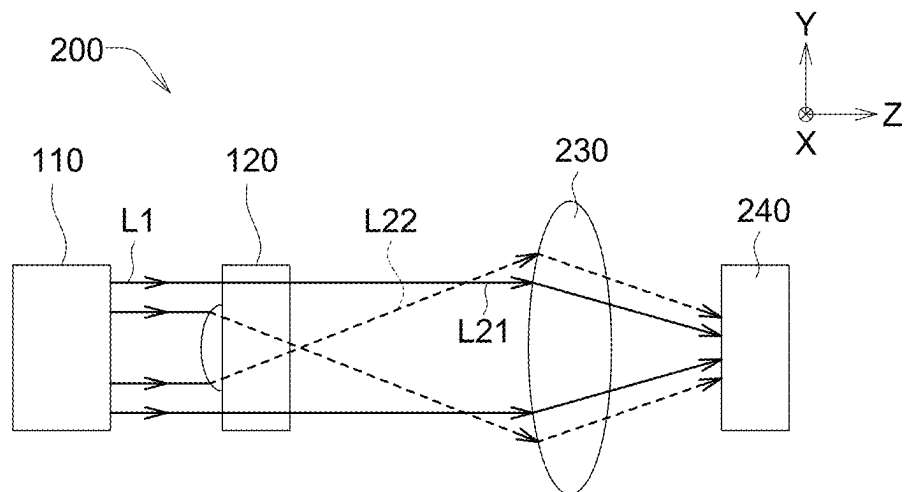


FIG. 6

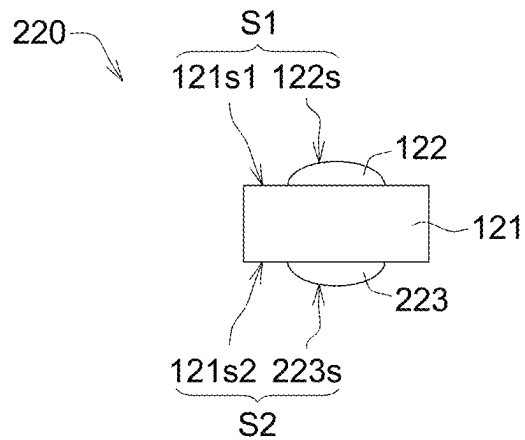


FIG. 7

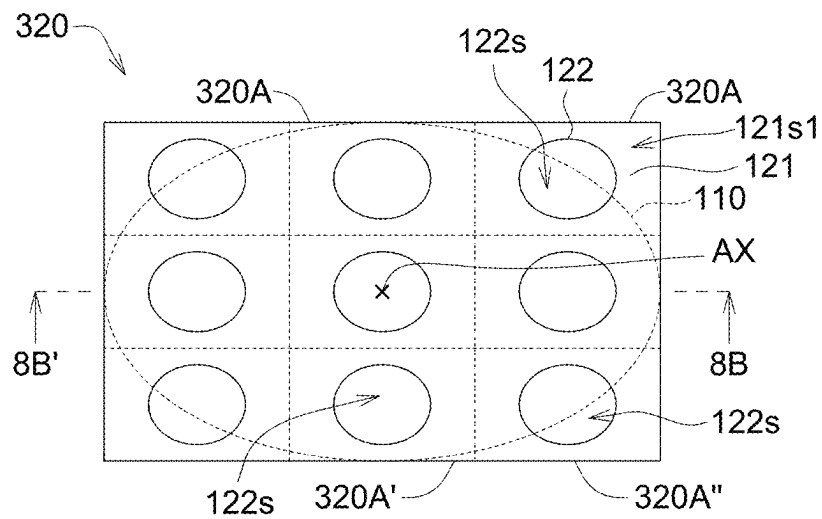


FIG. 8A

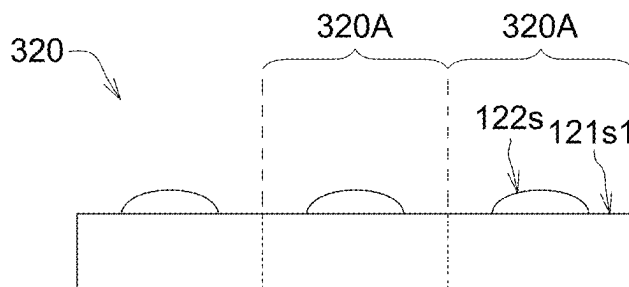


FIG. 8B

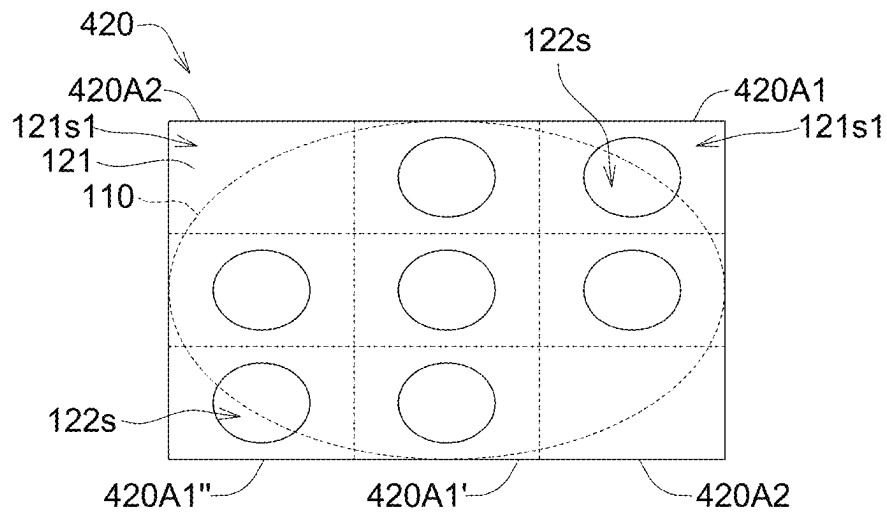


FIG. 9

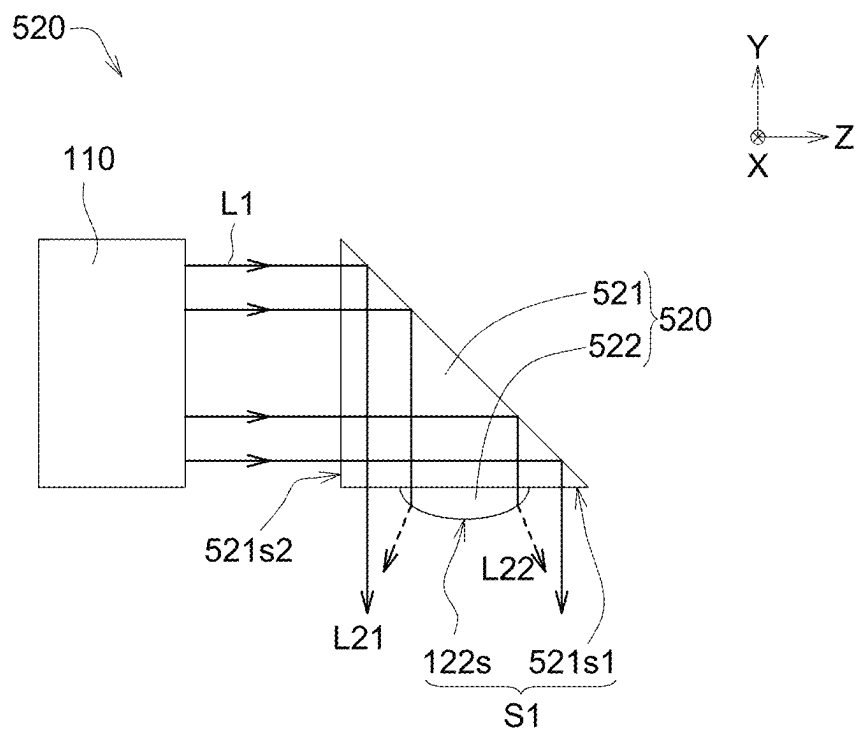
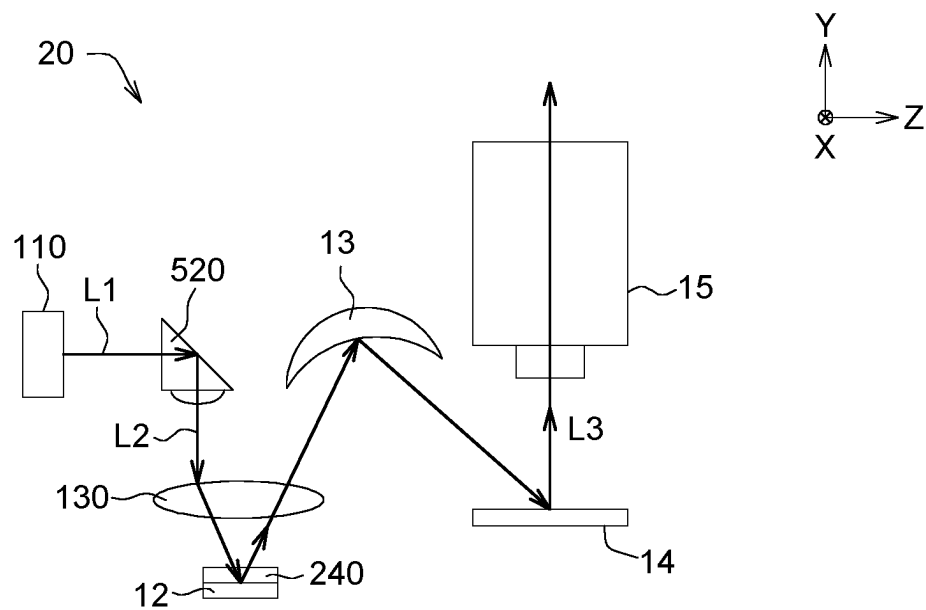
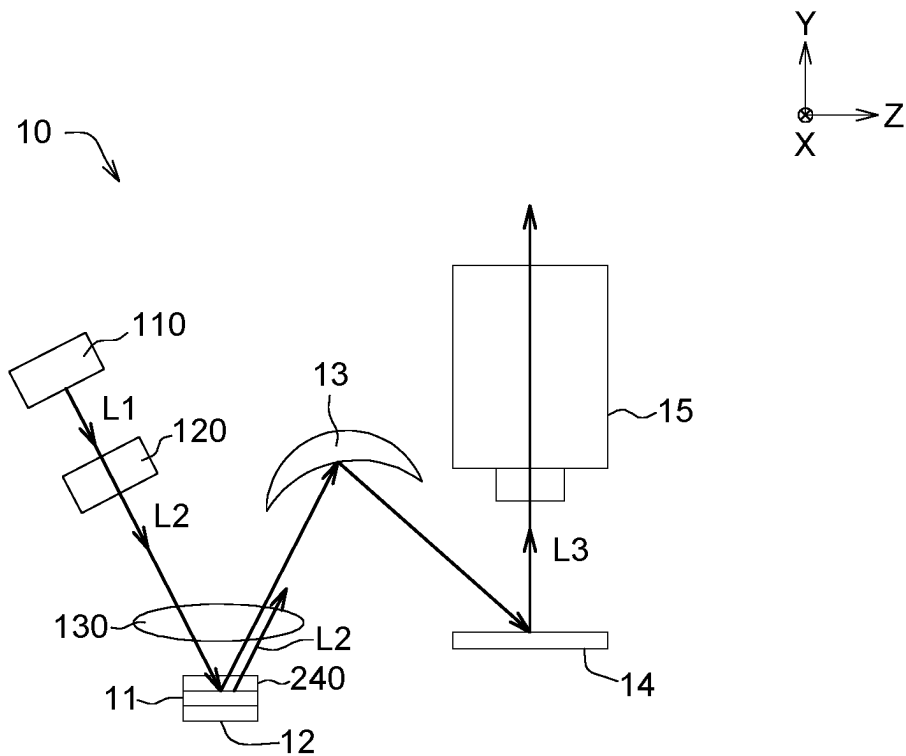


FIG. 10



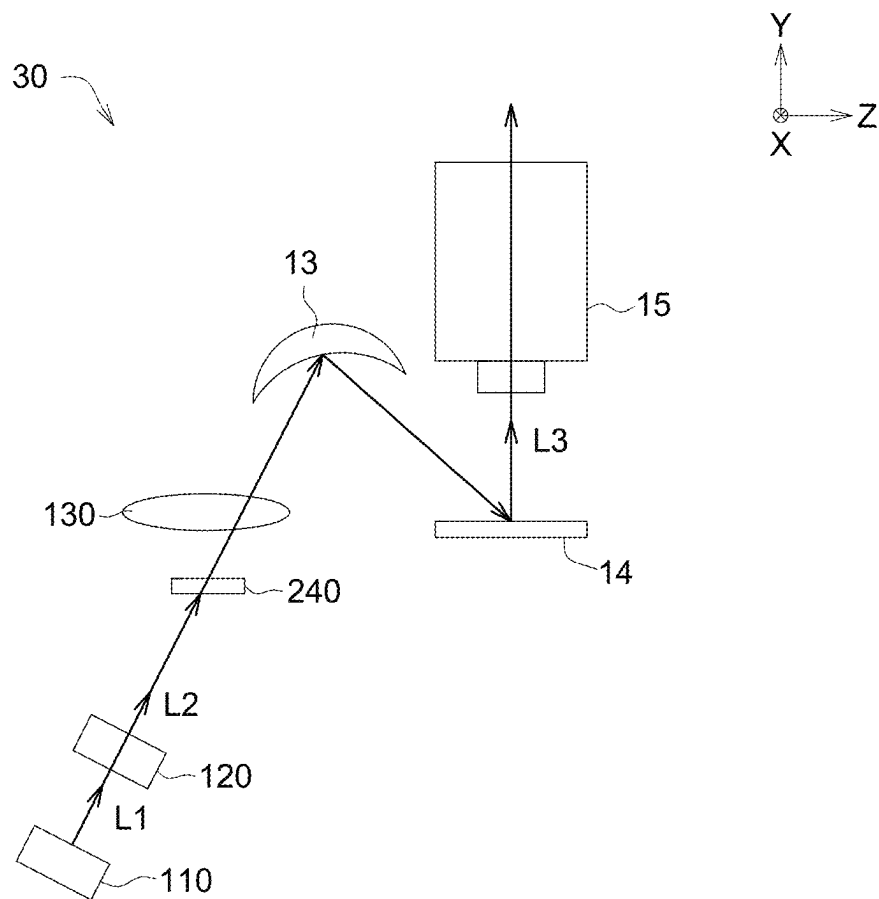


FIG. 13

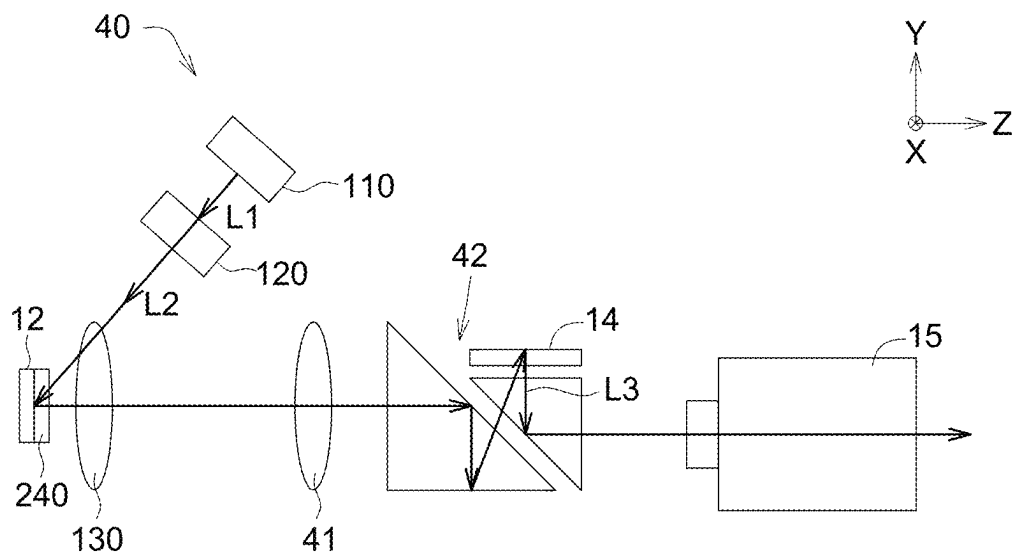


FIG. 14

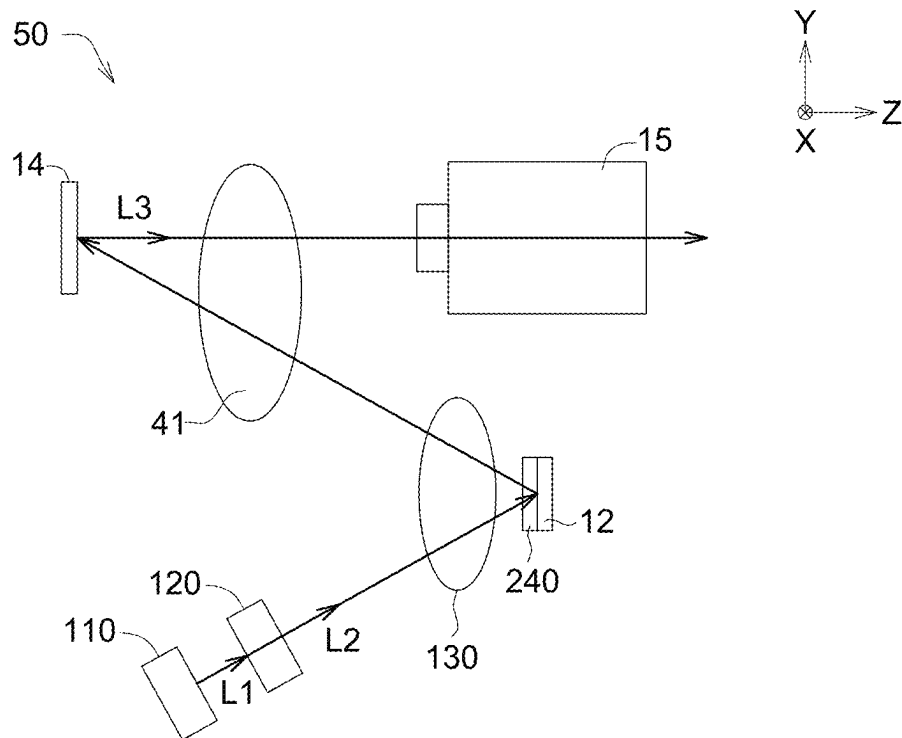


FIG. 15

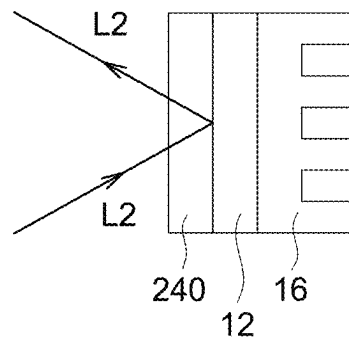


FIG. 16

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LIGHT SOURCE SYSTEM

This application claims the benefit of People's Republic of China application Serial No. 202311208559.9, filed on Sep. 19, 2023, the subject matter of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION**Field of the Invention**

The invention relates in general to a light source system.

Description of the Related Art

A conventional light source system provides a uniform illumination light. However, such uniform illumination light is not suitable for vehicle-mounted light source system. Therefore, proposing a light source system suitable for being mounted on a vehicle is one of the goals of those in this technical field.

SUMMARY OF THE INVENTION

According to an embodiment of the present invention, a light source system is provided. The light source system includes a first light source and a beam conversion element. The first light source is configured to emit a first light. The beam conversion element is configured to convert the first light into a condensed light and a diffused light. The beam conversion element has a light transmission surface, the light transmission surface includes a flat region and a curved-surface region, the flat region completely surrounds the curved-surface region, the flat region is configured to convert the first light into the condensed light, and the curved-surface region is configured to convert the first light into the diffused light.

The above and other aspects of the invention will become better understood with regard to the following detailed description of the preferred but non-limiting embodiment(s). The following description is made with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a schematic diagram of a light source system 100 according to an embodiment of the present invention;

FIG. 2 illustrates a schematic diagram of a beam conversion element 120 of the light source system 100 in FIG. 1;

FIG. 3A illustrates a schematic diagram of the beam conversion element 120 of the light source system 100 in FIG. 2;

FIG. 3B illustrates a schematic diagram of a light pattern P obtained by using the beam conversion element 120 in FIG. 3A;

FIG. 3C illustrates a schematic diagram of a relationship between the luminous angle and the flux of the light pattern P in X-axis in FIG. 3B;

FIG. 4A illustrates a schematic diagram of a beam conversion element 120' according to another embodiment of the present invention;

FIG. 4B illustrates a schematic diagram of a light pattern P obtained by using the beam conversion element 120' in FIG. 4A;

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FIG. 4C illustrates a schematic diagram of a relationship between the luminous angle and the flux of the light pattern P in X-axis in FIG. 4B;

FIG. 5 illustrates a schematic diagram of a beam conversion element 120" according to another embodiment of the present invention;

FIG. 6 is a schematic diagram of a light source system 200 according to another embodiment of the present invention;

FIG. 7 illustrates a schematic diagram of a beam conversion element 220 according to another embodiment of the present invention;

FIG. 8A illustrates a schematic diagram of a beam conversion element 320 according to another embodiment of the present invention;

FIG. 8B illustrates a schematic diagram of a cross-sectional view of the beam conversion element 320 in FIG. 8A along the direction 8B-8B';

FIG. 9 illustrates a schematic diagram of a beam conversion element 420 according to another embodiment of the present invention;

FIG. 10 illustrates a schematic diagram of a beam conversion element 520 according to another embodiment of the present invention;

FIG. 11 illustrates a schematic diagram of a light source system 10 according to another embodiment of the present invention;

FIG. 12 illustrates a schematic diagram of a light source system 20 according to another embodiment of the present invention;

FIG. 13 illustrates a schematic diagram of a light source system 30 according to another embodiment of the present invention;

FIG. 14 illustrates a schematic diagram of a light source system 40 according to another embodiment of the present invention;

FIG. 15 illustrates a schematic diagram of a light source system 50 according to another embodiment of the present invention; and

FIG. 16 illustrates a schematic diagram of a heat dissipation sink 16 disposed on a wavelength conversion element of a light source system according to another embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

Referring to FIGS. 1 to 4C, FIG. 1 illustrates a schematic diagram of a light source system 100 according to an embodiment of the present invention, FIG. 2 illustrates a schematic diagram of a beam conversion element 120 of the light source system 100 in FIG. 1, FIG. 3A illustrates a schematic diagram of the beam conversion element 120 of the light source system 100 in FIG. 2, FIG. 3B illustrates a schematic diagram of a light pattern P obtained by using the beam conversion element 120 in FIG. 3A, FIG. 3C illustrates a schematic diagram of a relationship between the luminous angle and the flux of the light pattern P in X-axis in FIG. 3B, FIG. 4A illustrates a schematic diagram of a beam conversion element 120' according to another embodiment of the present invention, FIG. 4B illustrates a schematic diagram of a light pattern P obtained by using the beam conversion element 120' in FIG. 4A, and FIG. 4C illustrates a schematic diagram of a relationship between the luminous angle and the flux of the light pattern P in X-axis in FIG. 4B.

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The light source system **100** may be applied to a vehicle-mounted lighting system, such as a car light, which may emit an illumination light to illuminate a scene ahead and/or a road surface.

As illustrated in FIG. 1, the light source system **100** includes a first light source **110** and a beam conversion element **120**. The first light source **110** is configured to emit a first light **L1**. The beam conversion element **120** is configured to convert the first light **L1** into a condensed light **L21** and a diffused light **L22**. The beam conversion element **120** has a light transmission surface **S1**, wherein the light transmission surface **S1** includes a flat region **121s1** and a curved-surface region **122s**, wherein the flat region **121s1** completely surrounds (for example, 360 degrees) the curved-surface region **122s**. The flat region **121s1** is configured to convert the first light **L1** into the condensed light **L21**. The curved-surface region **122s** is configured to convert the first light **L1** into the diffused light **L22**. In the present embodiment, the condensed light **L21** may enhance the intensity of the illumination light (for example, a mixed light of the condensed light **L21** and the diffused light **L22**) of the light source system **100**, and the diffused light **L22** may increase the uniformity of the illumination light of the light source system **100**, so that the illumination light provides a light pattern with a brighter brightness in a middle portion and a more uniform light intensity distribution in an edge portion.

As illustrated in FIG. 1, the light transmission surface **S1** is, for example, a light-incident surface of the beam conversion element **120**, but it may also be a light-exit surface of the beam conversion element **120**.

As illustrated in FIG. 1, the first light **L1**, the condensed light **L21** and the diffused light **L22** have the same wavelength. In other words, the beam conversion element **120** does not change the wavelength of the first light **L1**. The first light source **110** is, for example, a laser light, and the first light **L1** is, for example, a collimated light. However, the embodiment of the present invention does not limit the type of the first light source **110**. As illustrated in FIGS. 1 and 2, the first light source **110** has a light-exit surface **110s** that overlaps the entirety of the curved-surface region **122s** along a direction of an optical axis **AX** of the first light **L1**, but it may also overlap a part of the curved-surface region **122s**. Through an overlapping area between the light-exit surface **110s** and the curved-surface region **122s** along the optical axis **AX** of the first light **L1**, the light pattern and/or the light intensity distribution mode of the light emitted by the light source system **100** may be changed.

As illustrated in FIG. 2, in the present embodiment, the shape (e.g., top view shape) of the curved-surface region **122s** of the beam conversion element **120** is an ellipse. In other embodiment, the shape of the curved-surface region **122s** of the beam conversion element **120** may be a circle, a polygon (for example, a triangle, a quadrilateral, a pentagon, etc.) or an arbitrary shape constituted by a straight line, a curved line or a combination thereof. As long as the technical effect may be achieved, the shape of the curved-surface region **122s** is not limited by the embodiment of the present invention. In addition, the light pattern of the illumination light projected by the light source system **100** may change according to the geometric shape of the curved-surface region **122s**. For example, the light pattern of the illumination light projected by the light source system **100** is similar to the geometric shape of the curved-surface region **122s**.

As illustrated in FIG. 1, the beam conversion element **120** includes a first portion **121** and a second portion **122**,

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wherein the second portion **122** is connected to the first portion **121**. The first portion **121** has the aforementioned flat region **121s1**, and the second portion **122** has the aforementioned curved-surface region **122s**. The second portion **122** protrudes relative to the flat region **121s1** of the first portion **121**. In addition, the entirety of the beam conversion element **120** is, for example, an integrally formed structure (that is, the first portion **121** and the second portion **122** are integrally formed in one piece), and it may be formed of a material, for example, plastic or glass.

As illustrated in FIG. 1, the curved-surface region **122s** may protrude relative to the flat region **121s1**. As long as the light condensing effect may be achieved, the embodiment of the present invention does not limit the geometric form and/or the curvature distribution of the curved-surface region **122s**.

The beam conversion element **120** in the embodiment of the present invention does not require all or almost of the first light **L1** to be converted into the diffused light **L22**, and therefore it is not an "optical homogenizer". In an embodiment, different from the optical homogenizer, a flat proportion of the area of the flat region **121s1** of the beam conversion element **120** to the area of the light transmission surface **S1** is substantially equal to or higher than 10%. In an embodiment, the flat proportion of the area of the flat region **121s1** to the area of the light transmission surface **S1** is any real number between 10% and 90%. The higher the flat proportion is, the stronger the light intensity of the condensed light **L21** is.

As illustrated in FIGS. 3A and 4A, the flat proportion of the beam conversion element **120** in FIG. 3A is larger than that of the beam conversion element **120'** in FIG. 4A. As a result, as illustrated in FIGS. 3B and 3C, a ratio of the illumination (generated by the condensed light **L21**) of a middle region (the region A as illustrated in FIG. 3B) of the illumination light to the illumination (generated by the diffused light **L22**) of an edge region (the region B other than the region A as illustrated in FIG. 3B) of the illumination light which obtained by the beam conversion element **120** of the light source system **100** is greater. As illustrated in FIG. 4C, a ratio of the illumination (generated by the condensed light **L21**) of the middle region (the region A as illustrated in FIG. 4B) of the illumination light to the illumination (generated by the diffused light **L22**) of the edge region (the region B other than the region A as illustrated in FIG. 4B) of the illumination light which obtained by the beam conversion element **120'** of the light source system **100** is less. It may be seen that the higher the flat proportion is, the stronger the light intensity of the condensed light **L21** relative to the edge portion is. The luminous angle in FIGS. 3C and 4C is, for example, an angle included between the emitted-light of the light source system **100** and the optical axis **AX** (illustrated in FIG. 1).

In an embodiment, a ratio of the illumination at a center point (for example, point a in FIGS. 3C and 4C) of the light pattern to the illumination at an edge (for example, point b in FIGS. 3C and 4C) of the light pattern is **M**, wherein the ratio **M** may range between 3.5 and 17. Comparing the light pattern illumination distributions in FIGS. 3C and 4C, it may be seen that the ratio **M1** (for example, $M1=M1a/M1b$) obtained by the beam conversion element **120** of the light source system **100** is greater than the ratio **M2** (for example, $M2=M2a/M2b$) obtained by the beam conversion element **120'** of the light source system **100**. In other words, the greater the flat proportion is, the greater the ratio is.

Referring to FIG. 5, FIG. 5 illustrates a schematic diagram of a beam conversion element **120''** according to another

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embodiment of the present invention. The beam conversion element **120"** has a light transmission surface **S1"**, wherein the light transmission surface **S1"** includes a flat region **121s1** and a curved-surface region **122s"**. The flat region **121s1** completely surrounds (for example, 360 degrees) the curved-surface region **122s"**. The flat region **121s1** is configured to convert the first light **L1** into the condensed light **L21**, and the curved-surface region **122s** is configured to convert the first light **L1** into the diffused light **L22**. The beam conversion element **120"** includes a first portion **121** and a second portion **122"**, wherein the first portion **121** has the aforementioned flat region **121s1**, and the second portion **122"** has the aforementioned curved-surface region **122s"**. In the present embodiment, the second portion **122"** is recessed relative to the flat region **121s1** of the first portion **121**. The curved-surface region **122s"** of the second portion **122"** is recessed relative to the flat region **121s1** of the first portion **121**. The light transmission surface **S1"** is, for example, the light-incident surface of the beam conversion element **120"**, but it may also be the light-exit surface. In addition, the entirety of the beam conversion element **120"** is, for example, an integrally formed structure (that is, the first portion **121** and the second portion **122"** are integrally formed into one piece), and it may be formed of a material including, for example, plastic or glass.

Referring to FIG. 6, FIG. 6 is a schematic diagram of a light source system **200** according to another embodiment of the present invention. The light source system **200** includes the first light source **110**, the beam conversion element **120**, a lens **230** and a wavelength conversion element **240**. The first light source **110** is configured to emit the first light **L1**. The beam conversion element **120** is configured to convert the first light **L1** into condensed light **L21** and diffused light **L22**. The beam conversion element **120** has the light transmission surface **S1**. The light transmission surface **S1** includes the flat region **121s1** and the curved-surface region **122s**. The flat region **121s1** completely surrounds the curved-surface region **122s**. The flat region **121s1** is configured to convert the first light **L1** into the condensed light **L21**. The curved-surface region **122s** is configured to convert the first light **L1** into the diffused light **L22**. In the present embodiment, the condensed light **L21** may enhance the intensity of the illumination light (for example, the mixed light of the condensed light **L21** and the diffused light **L22**) of the light source system **200**, and the diffused light **L22** may increase the uniformity of the illumination light of the light source system **200**, so that the illumination light provides a light pattern with a brighter brightness in the middle portion and the more uniform light intensity distribution in the edge portion.

As illustrated in FIG. 6, the beam conversion element **120** is disposed between the lens **230** and the first light source **110**. The wavelength conversion element **240** is disposed on a downstream position of the lens **230**. The condensed light **L21** and the diffused light **L22** have a first wavelength.

As illustrated in FIG. 6, the wavelength conversion element **240** is disposed on the downstream position of the lens **230** and is configured to convert the first wavelength into a second wavelength that is different from the first wavelength. In an embodiment, the first wavelength is, for example, a blue light wavelength, and the second wavelength is, for example, a white light wavelength or a yellow light wavelength. In an embodiment, the wavelength conversion element **240** includes phosphors which may convert or change the wavelength of light.

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In addition, the structure of the beam conversion element is not limited by the aforementioned embodiments, and other structures of the beam conversion element will be further illustrated below.

Referring to FIG. 7, FIG. 7 illustrates a schematic diagram of a beam conversion element **220** according to another embodiment of the present invention. The beam conversion element **220** has the light transmission surface **S1** and a light transmission surfaces **S2** opposite to the light transmission surface **S1**. In the present embodiment, the light transmission surface **S1** may be one of the light-incident surface and the light-exit surface of the beam conversion element **220**, and the light transmission surface **S2** may be the other one of the light-incident surface and the light-exit surface of the beam conversion element **220**. The light transmission surface **S1** includes the flat region **121s1** and the curved-surface region **122s**, wherein the flat region **121s1** is configured to convert the first light **L1** into the condensed light **L21**, and the curved-surface region **122s** is configured to convert the first light **L1** into the diffused light **L22**. Similarly, the light transmission surface **S2** includes the flat region **121s2** and a curved-surface region **223s**, wherein the flat region **121s2** is configured to convert the first light **L1** into the condensed light **L21**, and the curved-surface region **223s** is configured to convert the first light **L1** into the diffused light **L22**.

As illustrated in FIG. 7, the beam conversion element **220** includes the first portion **121**, the second portion **122** and the third portion **223**. The second portion **122** and the third portion **223** are connected to two opposite sides of the first portion **121** respectively. For example, the first portion **121** has the aforementioned flat regions **121s1** and **121s2**, the second portion **122** has the aforementioned curved-surface region **122s**, and the third portion **223** has the aforementioned curved-surface region **223s**. The second portion **122** protrudes (for example, the curvature is defined as positive (+)) relative to the flat region **121s1** of the first portion **121**, and the third portion **223** protrudes relative to the flat region **121s2** of the first portion **121**. In another embodiment, the second portion **122** may be recessed (for example, the curvature is defined as negative (-)) relative to the flat region **121s1** of the first portion **121**, while the third portion **223** may be recessed relative to the flat region **121s2** of the first portion **121**. In other embodiment, the second portion **122** may be protruded or recessed relative to the flat region **121s1** of the first portion **121**, and the third portion **223** may be recessed or protruded relative to the flat region **121s2** of the first portion **121**. In addition, the entirety of the beam conversion element **220** is, for example, an integrally formed structure, and it may be formed of a material including, for example, plastic or glass.

Referring to FIGS. 8A and 8B, FIG. 8A illustrates a schematic diagram of a beam conversion element **320** according to another embodiment of the present invention, and FIG. 8B illustrates a schematic diagram of a cross-sectional view of the beam conversion element **320** in FIG. 8A along the direction 8B-8B'.

As illustrated in FIGS. 8A and 8B, the beam conversion element **320** includes a plurality of adjustment units **320A**. Each adjustment unit **320A** includes the aforementioned flat region **121s1** and curved-surface region **122s**, or includes the aforementioned first portion **121** and second portion **122**. A plurality of the adjustment units **320A** may be connected to each other and arranged in an nxm array, wherein n and m are positive integers equal to or greater than 1.

The dotted line range shown in FIG. 8A represents the overlapping range where the light-exit surface **110s** of the first light source **110** is projected on the beam conversion

element **320** in the direction of the optical axis **AX** of the first light **L1**. It may be seen from the dotted line range that the light-exit surface **110s** overlaps the entirety of one (for example, the curved-surface region **122s** of the adjustment unit **320A'**) of the curved-surface regions **122s**, and overlaps a portion of another (for example, the curved-surface region **122s** of the adjustment unit **320A''**) of the curved-surface regions **122s**. Through the overlapping range design or the position design of the light-exit surface **110s** and the beam conversion element **320**, the corresponding (or different) relationship curve between the luminous angle and the illumination may be obtained.

Although not illustrated, in another embodiment, the adjustment unit **320A** may include the flat region **121s1**, the flat region **121s2**, the curved-surface region **122s** and the curved-surface region **223s** in FIG. 7, or include the aforementioned first portion **121**, second portion **122** and third portion **223**.

Referring to FIG. 9, FIG. 9 illustrates a schematic diagram of a beam conversion element **420** according to another embodiment of the present invention. The beam conversion element **420** includes at least one adjustment unit **420A1** and at least one adjustment unit **420A2**, wherein the adjustment unit **420A1** includes the aforementioned flat region **121s1** and curved-surface region **122s**, or includes the aforementioned first portion **121** and second portion **122**, and the adjustment unit **420A2** includes the aforementioned flat surface region **121s1**, but does not include the curved-surface region **122s**, or includes the aforementioned first portion **121** but does not include the second portion **122**. A plurality of the adjustment units **420A** is arranged in an $n \times m$ array, wherein n and m are positive integers equal to or greater than 1.

Although not illustrated, in another embodiment, the adjustment unit **420A1** may include the flat region **121s1**, the flat region **121s2**, the curved-surface region **122s** and the curved-surface region **223s** in FIG. 7, or the aforementioned first portion **121**, second portion **122** and third portion **223**, and the adjustment unit **420A2** may include the flat regions **121s1** and **121s2** in FIG. 7 but does not include the curved-surface region **122s** and the curved-surface region **223s**, or include the aforementioned first portion **121** but does not include the aforementioned second portion **122** and third portion **223**.

The dotted line range shown in FIG. 9 represents the overlapping range where the light-exit surface **110s** of the first light source **110** is projected on the beam conversion element **420** along the direction of the optical axis **AX** of the first light **L1**. It may be seen from the dotted line range that the light-exit surface **110s** overlaps the entirety of one (for example, the curved-surface region **122s** of the adjustment unit **420A'**) of the curved-surface regions **122s**, overlaps a portion of another (for example, the curved-surface region **122s** of the adjustment unit **420A''**) of the curved-surface regions **122s**, and overlaps at least one portion of the flat region **121s1** of the adjustment unit **420A2**. Through the overlapping range design or the position design of the light-exit surface **110s** and the beam conversion element **420**, the corresponding (or different) relationship curve between the luminous angle and the illumination may be obtained.

Referring to FIG. 10, FIG. 10 illustrates a schematic diagram of a beam conversion element **520** according to another embodiment of the present invention. The light beam conversion element **520** includes a prism **521** and a curved-surface portion **522**, and the curved-surface portion **522** is disposed on the prism **521**. The beam conversion

element **520** has the light transmission surface **S1** and a light-incident surface **521s2**. The prism **521** has the light-incident surface **521s2** configured to receive the first light **L1**. The light transmission surface **S1** is, for example, the light-exit surface and includes a flat region **521s1** and the curved-surface region **122s**. The prism **521** has the flat region **521s1**, and the curved-surface portion **522** has the curved-surface region **122s**. The flat region **521s1** is used for converting the first light **L1** into the condensed light **L21**, and the curved-surface region **122s** is configured to convert the first light **L1** into the diffused light **L22**. In an embodiment, the entirety of the beam conversion element **520** is, for example, an integrally formed structure, and it may be formed of a material including, for example, plastic or glass.

Referring to FIG. 11, FIG. 11 illustrates a schematic diagram of a light source system **10** according to another embodiment of the present invention. The light source system **10** includes the first light source **110**, the beam conversion element **120**, a lens **130**, the wavelength conversion element **240**, a second light source **11**, a reflective element **12**, a reflective element **13**, a DMD (Digital Micro-mirror Device) **14** and a projection lens **15**.

As illustrated in FIG. 11, the first light source **110** is configured to emit the first light **L1**. The beam conversion element **120** is configured to convert the first light **L1** into condensed light **L21** (not illustrated in FIG. 11) and diffused light **L22** (not illustrated in FIG. 11). The condensed light **L21** and diffused light **L22** are mixed into an illumination light **L2**. The wavelength conversion element **240** is disposed between the second light source **11** and the lens **130** and is located at downstream position of the beam conversion element **120**, and the lens **130** is disposed on a downstream position of the wavelength conversion element **240**. The second light source **11** is disposed between the wavelength conversion element **240** and the reflective element **12**. In the present embodiment, the illumination light **L2** is projected to the front of the light source system **10** or a road surface through an optical path, wherein the optical path travels through the lens **130**, the wavelength conversion element **240**, the second light source **11**, the reflective element **12**, the second light source **11**, the wavelength conversion element **240**, the lens **130**, the reflective element **13**, the DMD **14** and projection lens **15**.

The second light source **11** is, for example, an unpackaged die, which has light-transmissive property and may allow light to travel through. The second light source **11** may emit the second light **L2** with the first wavelength. The wavelength conversion element **240** converts the first wavelength of the second light **L2** into a second wavelength. The second light **L2** of the second light source **11** may increase the intensity of the illumination light emitted from the light source system **10**. The reflective element **12** and/or the reflective element **13** are, for example, reflective layers or reflective mirrors. The DMD **14** may convert the illumination light **L2** into a patterned light **L3**, and the pattern of the patterned light **L3** may be changed by the DMD **14**.

Referring to FIG. 12, FIG. 12 illustrates a schematic diagram of a light source system **20** according to another embodiment of the present invention. The light source system **20** includes the first light source **110**, the beam conversion element **520**, the lens **130**, the wavelength conversion element **240**, the reflective element **12**, the reflective element **13**, the DMD **14** and the projection lens **15**.

As illustrated in FIG. 12, the first light source **110** is configured to emit the first light **L1**. The beam conversion element **520** is configured to convert the first light **L1** into condensed light **L21** (not illustrated in FIG. 12) and the

diffused light L22 (not illustrated in FIG. 12). The condensed light L21 and the diffused light L22 are mixed into the illumination light L2. The wavelength conversion element 240 is disposed between the reflective element 12 and the lens 130 and is located at the downstream position of the beam conversion element 120. The lens 130 is disposed on the downstream position of the wavelength conversion element 240. In the present embodiment, the illumination light L2 is projected to the front of the light source system 20 or the road surface through an optical path, wherein the optical path travels through the lens 130, the wavelength conversion element 240, the reflective element 12, the wavelength conversion element 240, the lens 130, the reflective element 13, the DMD 14 and projection lens 15.

In another embodiment, similar to FIG. 11, the second light source 11 may be disposed between the wavelength conversion element 240 and the reflective element 12.

Referring to FIG. 13, FIG. 13 illustrates a schematic diagram of a light source system 30 according to another embodiment of the present invention. The light source system 30 includes the first light source 110, the beam conversion element 120, the lens 130, the wavelength conversion element 240, the reflection element 13, the DMD 14 and the projection lens 15.

As illustrated in FIG. 13, the first light source 110 is configured to emit the first light L1. The beam conversion element 120 is configured to convert the first light L1 into the condensed light L21 (not illustrated in FIG. 12) and the diffused light L22 (not illustrated in FIG. 12). The condensed light L21 and the diffused light L22 are mixed into the illumination light L2. The wavelength conversion element 240 is disposed between the beam conversion element 120 and the lens 130 and is located at the downstream position of the beam conversion element 120. The lens 130 is disposed on the downstream position of the wavelength conversion element 240. In this embodiment, the illumination light L2 is projected to the front of the light source system 30 or the road surface through an optical path, wherein the optical path travels through the wavelength conversion element 240, the lens 130, the reflective element 13, the DMD 14 and the projection lens 15.

Referring to FIG. 14, FIG. 14 illustrates a schematic diagram of a light source system 40 according to another embodiment of the present invention. The light source system 40 includes the first light source 110, the beam conversion element 120, the lens 130, the wavelength conversion element 240, the reflection element 12, a lens 41, a prism assembly 42, a DMD 14 and a projection lens 15.

As illustrated in FIG. 14, the first light source 110 is configured to emit the first light L1. The beam conversion element 120 is configured to convert the first light L1 into the condensed light L21 (not illustrated in FIG. 14) and the diffused light L22 (not illustrated in FIG. 14). The condensed light L21 and the diffused light L22 are mixed into the illumination light L2. The wavelength conversion element 240 is disposed between the reflective element 12 and the lens 130 and is located at the downstream position of the beam conversion element 120. The lens 130 is disposed on the downstream position of the wavelength conversion element 240. In the present embodiment, the illumination light L2 is projected to the front of the light source system 40 or the road surface through an optical path, wherein the optical path travels through the lens 130, the wavelength conversion element 240, the reflective element 12, the wavelength conversion element 240, the lens 130, the lens 41, the prism assembly 42, the DMD 14 and the projection lens 15.

Referring to FIG. 15, FIG. 15 illustrates a schematic diagram of a light source system 50 according to another embodiment of the present invention. The light source system 50 includes the first light source 110, the beam conversion element 120, the lens 130, the wavelength conversion element 240, the reflection element 12, the lens 41, the DMD 14 and the projection lens 15.

As illustrated in FIG. 15, the first light source 110 is configured to emit the first light L1. The beam conversion element 120 is configured to convert the first light L1 into the condensed light L21 (not illustrated in FIG. 14) and the diffused light L22 (not illustrated in FIG. 14). The condensed light L21 and the diffused light L22 are mixed into illumination light L2. The wavelength conversion element 240 is disposed between the reflective element 12 and the lens 130 and is located at the downstream position of the beam conversion element 120. The lens 130 is disposed on the downstream position of the wavelength conversion element 240. In the present embodiment, the illumination light L2 is projected to the front of the light source system 50 or the road surface through an optical path, wherein the optical path travels through the lens 130, the wavelength conversion element 240, the reflective element 12, the wavelength conversion element 240, the lens 41, and the DMD 14, the lens 41 and projection lens 15.

In an embodiment, the wavelength conversion element of any embodiment herein is rotatably disposed on the downstream position of the beam conversion element. In another embodiment, the wavelength conversion element of any embodiment herein may be fixedly disposed on the downstream position of the beam conversion element.

Referring to FIG. 16, FIG. 16 illustrates a schematic diagram of a heat dissipation sink 16 disposed on a wavelength conversion element of a light source system according to another embodiment of the present invention. The light source system of any of the aforementioned embodiments may further include the heat dissipation sink 16 disposed adjacent to the wavelength conversion element 240. For example, the reflective element 12 is connected to the wavelength conversion element 240. Alternatively, the heat dissipation sink 16 may be directly connected or be connected to the wavelength conversion element 240 of the light source system of any of the aforementioned embodiments through thermally conductive glue.

In addition, the beam conversion element of the light source system of any of the aforementioned embodiments may be replaced by the beam conversion element of another embodiment.

To sum up, the light source system includes a first light source and a beam conversion element. The first light source emits a first light to the beam conversion element. The beam conversion element is transparent and may convert the first light into an illumination light (a mixed light of the condensed light and the diffused light), the illumination light provides a light pattern with a concentrated brightness in a middle portion and an uniform brightness in an edge portion. The beam conversion element has at least one light transmission surface, wherein the light transmission surface includes a flat region and a curved-surface region. The flat region surrounds the curved-surface region to achieve the aforementioned light pattern. In another embodiment, the light source system further includes a wavelength conversion element and a reflective element. The reflective element is disposed on a downstream position of the wavelength conversion element to reflect the illumination light. In other embodiments, the light source system further includes a second light source which is disposed between the reflective

element and the wavelength conversion element to provide the second light and increase the overall intensity of the illumination light of the light source system.

While the invention has been described by way of example and in terms of the preferred embodiment (s), it is to be understood that the invention is not limited thereto. Based on the technical features embodiments of the present invention, a person ordinarily skilled in the art will be able to make various modifications and similar arrangements and procedures without breaching the spirit and scope of protection of the invention. Therefore, the scope of protection of the present invention should be accorded with what is defined in the appended claims.

What is claimed is:

1. A light source system, comprising:
a first light source configured to emit a first light; and
a beam conversion element configured to convert the first light into a condensed light and a diffused light;
wherein the beam conversion element has a light transmission surface, the light transmission surface comprises a flat region and a curved-surface region, the flat region completely surrounds the curved-surface region, the flat region is configured to convert the first light into the condensed light, and the curved-surface region is configured to convert the first light into the diffused light;
wherein the first light, the condensed light and the diffused light have the same wavelength.

2. The light source system according to claim 1, wherein the curved-surface region is protruded or recessed relative to the flat region.

3. The light source system according to claim 1, wherein the flat region and the curved-surface region form an adjustment unit, the beam conversion element comprises a plurality of the adjustment units, and the adjustment units are arranged in an $n \times m$ array, where n and m are positive integers equal to or greater than 1.

4. The light source system according to claim 3, wherein the first light source has a light-exit surface, the light-exit surface overlaps the entirety of one of the curved-surface regions and overlaps a portion of another of the curved-surface regions along a direction of an optical axis of the first light.

5. The light source system according to claim 1, wherein the beam conversion element comprises a plurality of adjustment units, one of the adjustment units comprises the flat region and the curved-surface region, and another of the adjustment units comprises the flat region but does not comprise the curved-surface region.

6. The light source system according to claim 1, wherein the flat region has an area accounting for more than 10% of an area of the light transmission surface.

7. The light source system according to claim 1, wherein the flat region has an area accounting for 10% to 90% of an area of the light transmission surface.

8. The light source system according to claim 1, wherein the first light source has a light-exit surface, the light-exit surface overlaps the entirety of the curved-surface region along a direction of an optical axis of the first light.

9. The light source system according to claim 1, further comprising:

a lens;
wherein the beam conversion element is disposed between the lens and the first light source.

10. The light source system according to claim 1, wherein the condensed light and the diffused light have a first wavelength, and the light source system further comprises:

a lens; and

a wavelength conversion element disposed on a downstream position of the lens and configured to convert the first wavelength into a second wavelength, wherein the second wavelength is different from the first wavelength;

wherein the beam conversion element is disposed between the lens and the first light source.

11. The light source system according to claim 1, further comprising:

a wavelength conversion element;

a lens; and

a second light source configured to emit a second light to the wavelength conversion element;

wherein the wavelength conversion element is disposed between the second light source and the lens.

12. The light source system according to claim 11, further comprising:

a reflective element;

wherein the second light source is disposed between the wavelength conversion element and the reflective element.

13. The light source system according to claim 1, wherein the beam conversion element comprises:

a prism; and

a curved-surface portion disposed on the prism and having the curved-surface region.

14. The light source system according to claim 13, wherein the prism and the curved-surface portion are integrally formed into one piece.

15. The light source system according to claim 13, wherein the prism has a light-incident surface and the flat region, the light-incident surface is configured to receive the first light, and the curved-surface portion has the curved-surface region.

16. The light source system according to claim 1, wherein the condensed light and the diffused light have a first wavelength, and the light source system further comprises:

a wavelength conversion element configured convert the first wavelength into a second wavelength, wherein the second wavelength is different from the first wavelength; and

a lens disposed on a downstream portion of the wavelength conversion element.

17. The light source system according to claim 1, wherein the condensed light and the diffused light have a first wavelength, and the light source system further comprises:

a wavelength conversion element rotatably disposed on a downstream position of the beam conversion element and configured to convert the first wavelength into a second wavelength, wherein the second wavelength is different from the first wavelength.

18. The light source system according to claim 1, wherein the condensed light and the diffused light have a first wavelength, and the light source system further comprises:

a wavelength conversion element fixedly disposed on a downstream position of the beam conversion element, and configured to convert the first wavelength into a second wavelength, wherein the second wavelength is different from the first wavelength; and

a heat dissipation sink connected to the wavelength conversion element.

19. The light source system according to claim 1, wherein the light beam conversion element has two opposite light transmission surfaces.

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20. The light source system according to claim **1**, wherein a shape of the curved-surface region is a circle, an ellipse or a polygon.

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